# Introduction to Medical Image Analysis:

# Segmentation of lungs from X-ray using basic image processing techniques

- Image segmentation is a process of partitioning an image into multiple distinctive regions (image segments, which are sets of pixels in 2D [or voxels in 3D]). In segmentation process, we assign a label to every pixel in an image in such way that the pixels having the same label share similar characteristics.
- In this tutorial, the task is to segment lungs by creating a binary image from a chest X-ray, as this is the simplest way to divide image into our object of interest (lung) and background (everything else). The tutorial will use segmentation algorithms (e.g. thesholding algorithms), image filtering algorithms, morhological operations from scikit-image (https://scikit-image.org/)
- the stack of chest X-rays from standard digital image dataset for Tuberculosis (created by The National Library of Medicine in Collaboration with The Department Of Health And Human Services). The data set consists of 80 x-rays for normal cases, and 53 cases with manifestation of tuberculosis.

- **Data set:**
  **Please use Montgomery County X-ray Set:**
  Local copy: /data/cdtshared/Cohort-2024/HDS-M03/MontgomerySet

  Online version (maybe slow): http://openi.nlm.nih.gov/imgs/collections/NLM-MontgomeryCXRSet.zip

- **Image files names:**
  Image file names are coded as MCUCXR_#####_0/1.png, where '0' represents the normal and '1' represents the abnormal lung with manifestation of tuberculosis.

- **Ground Truth:**
  The data set was also manually labelled under supervision of radiologist. The lung boundaries (left and right) are in binary image format and have the same file name as chest X-rays ( e.g. …/left/MCUCXR_#####_0/1.png or …/right/MCUCXR_#####_0/1.png).

## Preparing imaging data:

1. The original data set contains 138 x-ray images and two labels (separate label for left and right lung), however due to the limited time, please consider to use a subset of the images (e.g. normal) for a few cases only.
2. Load and visualise a few x-ray images and their ground truth segmentations (using e.g. matplot, see: https://matplotlib.org/stable/tutorials/introductory/images.html)
3. You may consider to resample them as the size of the original images could be up to 4892x4020 pixels.

## Supervised thresholding:

4. Create a binary mask of image applying a basic (and naive) threshold algorithm by using e.g. the mean value of pixel intensities.

You may want to display histogram of chest x-ray to better pick the value which could give us a reasonable segmentation.

## Unsupervised thresholding:

5. There have been multiple methods developed to estimate 'optimal' threshold value, e.g. https://en.wikipedia.org/wiki/Otsu's_method

## Further image processing:

6. Your segmentation may still not be perfect, therefore it could be good idea to use perform some filtering methods, morphological operations

## Assessment of segmentation and use of ground truth:

7. The CXR data has also ground truth segmentations created manually by a radiologist. How could you use it for validating your results?
8. Can you repeat this segmentation process for abnormal cases? Can your report quality of segmentation for a stack of normal and abnormal cases?

## Classification:

9. Calculate a set to features that are based on e.g. Gray-Level Co-Occurrence Matrix features and try to develop a classifier. Calculate on GLCM of the classes normal vs abnormal (TB cases). Use a scatter plot or any other suitable statistics to see if the feature statistic of the two classes is different.